

After harvest, straw needs a buyer and a baler

**A narrow window. Many fields.
Scattered machinery and demand.**

Farmers

Need a timely straw pickup option.

Operators

Need a clear dispatch queue.

Buyers

Need visible, traceable supply; this prototype does not establish actual demand.

One evidence-to-action workflow

Observe

Sentinel-2 field histories

Prioritize

Normalized burn risk score, not probability

Coordinate

Demo baler/buyer workflow

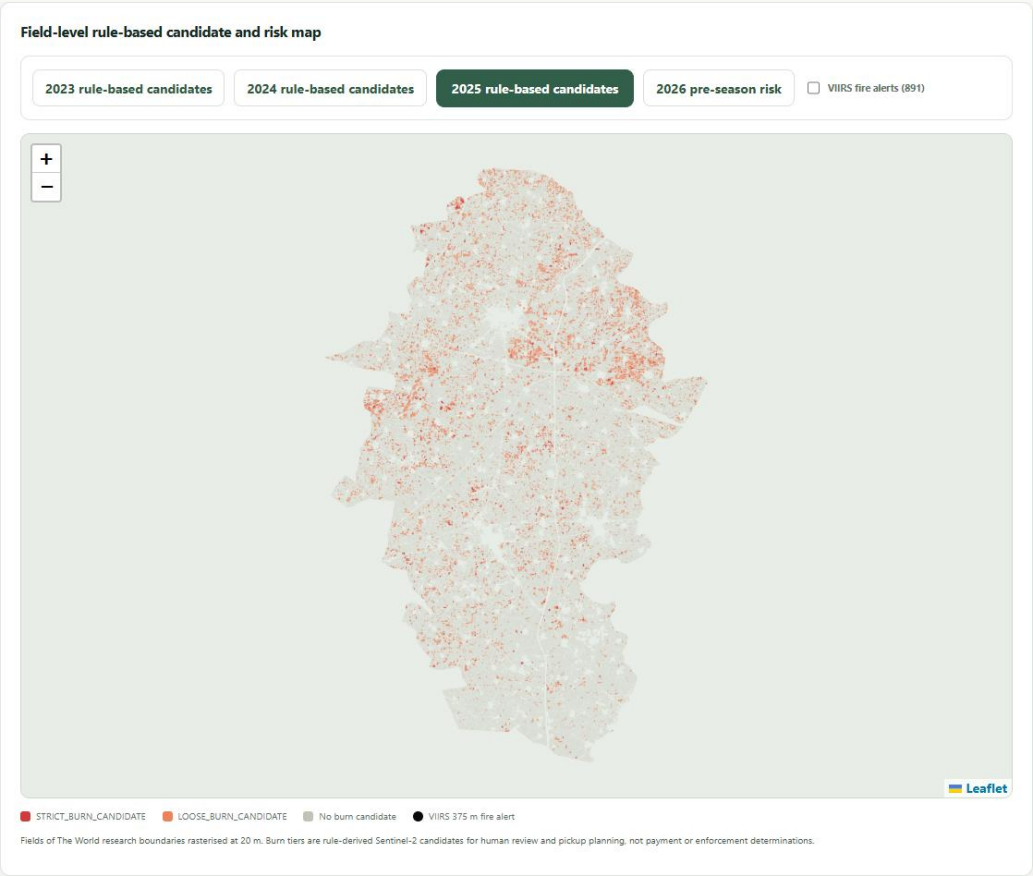
Record

Prototype QR metadata check

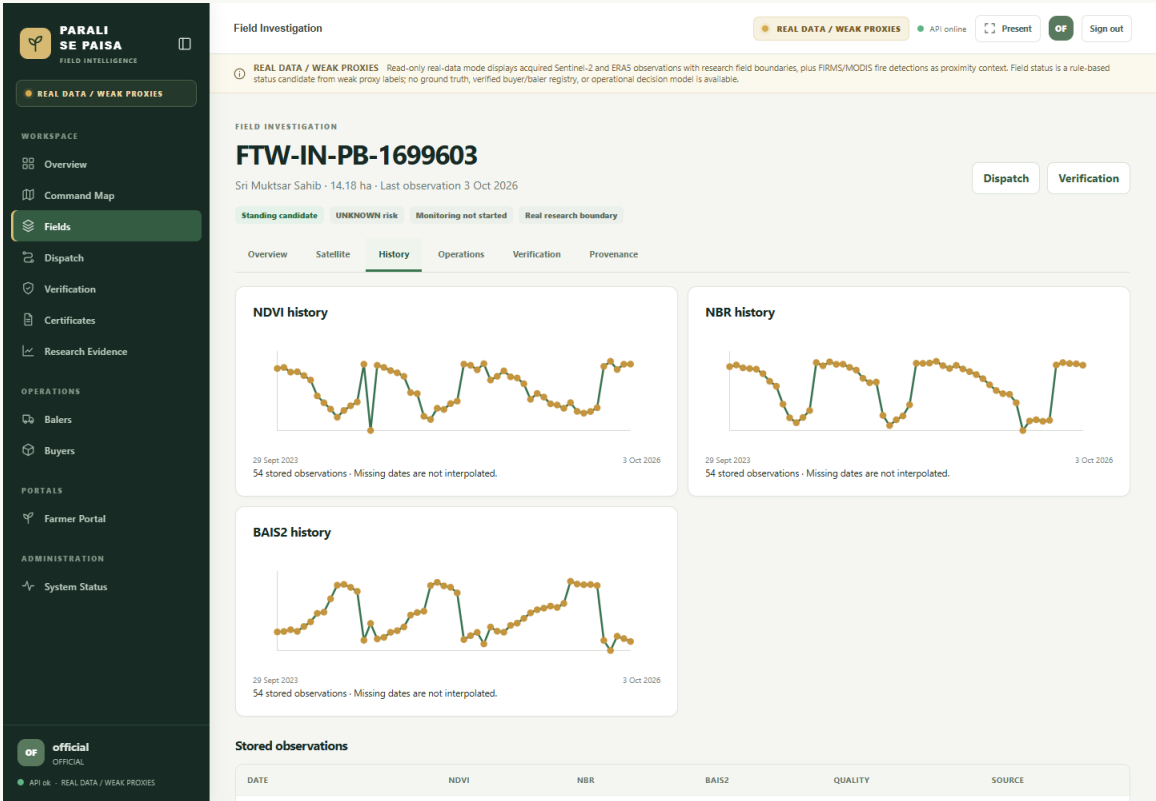
REAL DATA → RULE-BASED INTERPRETATION → DEMO/SYNTHETIC OPERATIONS → PROTOTYPE RECORD

Muktsar map and one field history, 2023–2026

74 Sentinel-2 scenes · 12.6 million field-date observations



REAL DATA · rule-based candidate map



REAL DATA · one field, stored observations

Source: fresh screenshots from isolated read-only real-data mode; 74 Sentinel-2 scenes, 12.6 million field-date observations; Fields of The World boundaries are model-derived.

Candidate area depends on the rule tier

LOOSE_BURN_CANDIDATE

2023: 21,121 ha | 2024: 10,566 ha | 2025: 22,941 ha. Full-revisit, exploratory rule-based area.

STRICT_BURN_CANDIDATE

2023: 649 ha | 2024: 1,022 ha | 2025: 2,269 ha. Not a trend: haze/cloud changes clean pre-event coverage.

All outputs are RULE-BASED BURN CANDIDATES. Human review labels completed: 0.

Optical candidates and thermal context answer different questions

8.8%

of 2025 strict optical candidates had a matched VIIRS active-fire detection within the defined window.

What the match means

About 9 in 10 strict optical burn candidates had no matched VIIRS active-fire detection within the defined matching window.

Interpretation

VIIRS is corroborative thermal context, not ground truth or a field-level confirmation.

A reported straw figure needs its source

12.1 lakh tonnes

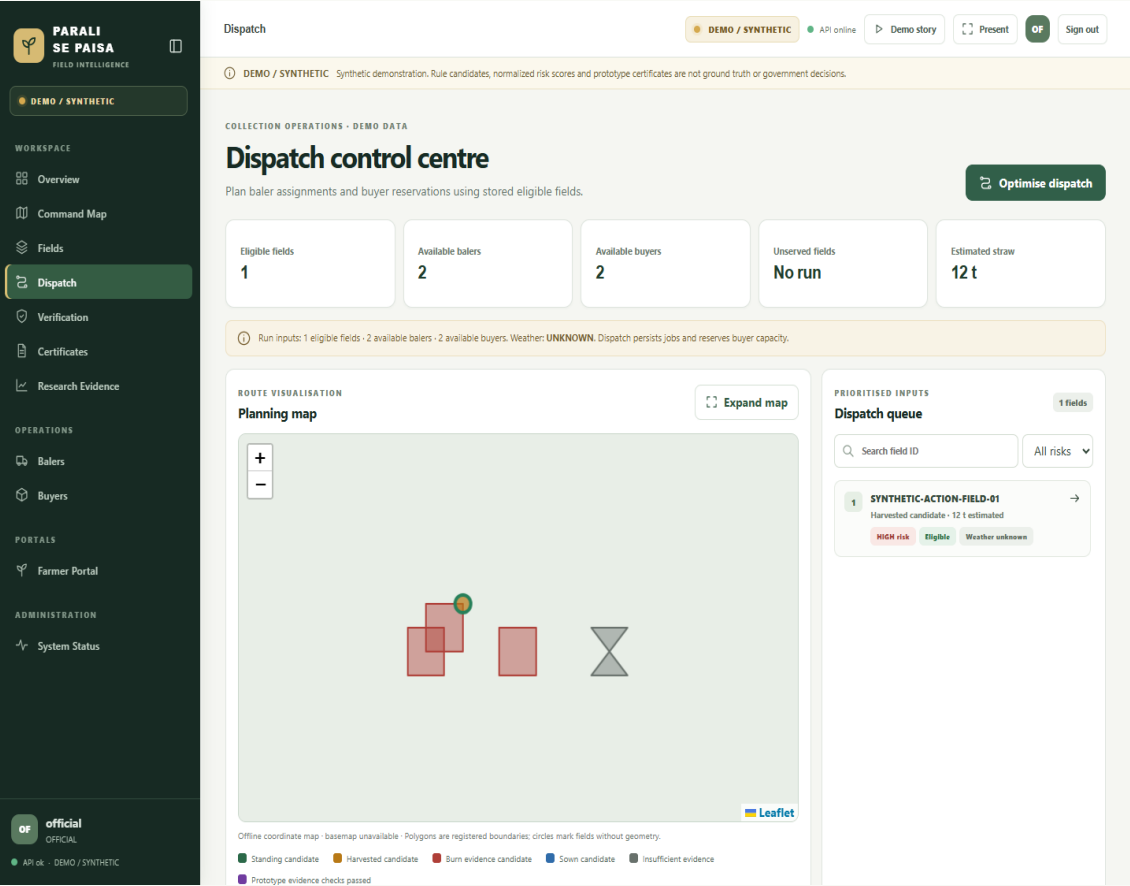
The Tribune reported this straw estimate. It is not an official statistic.

From estimate to action

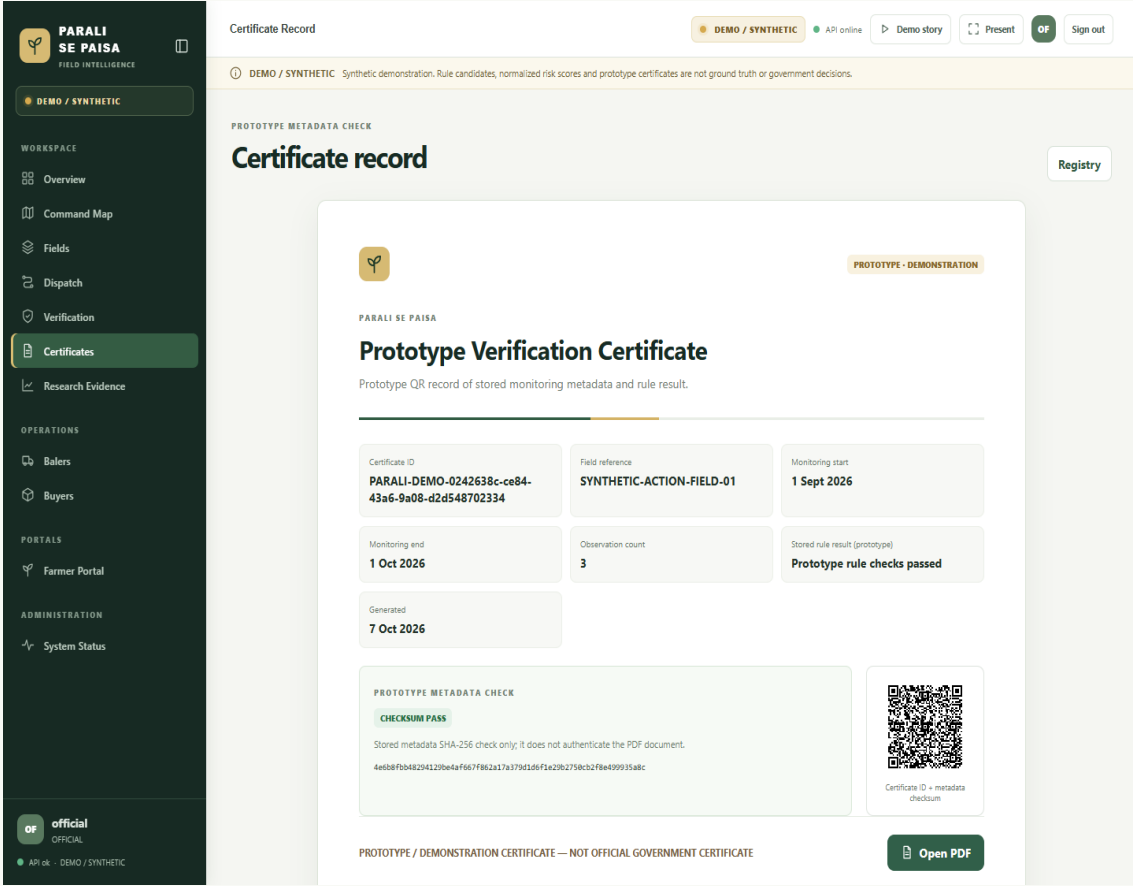
The product concept connects field evidence to straw pickup planning; actual supply and buyer demand still need field confirmation.

Attribution stays with the number wherever it appears: The Tribune (reference 11 in the concept note).

Dispatch is an operational prototype



DEMO/SYNTHETIC • route proxy



PROTOTYPE • QR metadata only

What the evidence can and cannot support

No human validation yet

0 completed blinded labels

No strict trend

Pre-event clarity varies by season

No ground truth claim

Thermal matches are corroborative context

No probability claim

Risk is normalized score, not probability

Pilot the evidence-to-pickup loop

Real satellite evidence → clear limits → a prototype path from straw to value.

1 · Inspect

Muktsar real map and one field history

2 · Demonstrate

Synthetic dispatch and buyer workflow

3 · Pilot

Independent labels and local partners